SE311 SPRING 2025-2026 / 18-03-2026 && 19-03-2026
LAB 04 – Command Pattern

In this lab, you will simulate a game environment where a **`GameCharacter`** moves within a 2D grid. To manage these movements flexibly, you will implement a system where every directional input is encapsulated as a command object, allowing for features like movement history and reversing actions.

You are required to implement four distinct commands that represent the directions of movement:

- **MoveUpCommand**: Increases the vertical position of the character.
- **MoveDownCommand**: Decreases the vertical position of the character.
- **MoveLeftCommand**: Decreases the horizontal position of the character.
- **MoveRightCommand**: Increases the horizontal position of the character.

When any of these commands are executed, the system should print a message indicating the action, such as "Moving Up!" or "Moving Left!".

User will decide on the movement which he/she is going to take. After that, he/she will press on the directional keys on the **`Keyboard`**. (You are simulating keyboard's directional keys on Java like **pressUpDirectionKey()** method for pressing Upper Directional Key.)

Keyboard will understand the key that user pressed on and will store them in its storage.

Also, this simulational Keyboard has a special key called "UNDOKEY" for undoing the last operation it performed. (User again will press this key with **pressUndoKey()** method.)

When you have finished with all of these, you will implement a macro move which is called "L-Shape Move" in this movement firstly, you will go up twice and go one-time right-hand side. This allows a character to jump forward and to the right-hand side in one action.

Create a main method and test all the commands in the system.